

A Benchmark and an Open Problem: Fractal-Basin Classification for the Planar Three-Body Problem

Aishwarya Das
Dirac Labs Inc.
aishwarya@diraclabs.com

Abstract

The escape basins of the gravitational three-body problem have fractal boundaries whose geometry constrains learnability. We release a benchmark of 10^6 labeled initial conditions on the Montgomery shape sphere (2D, at rest) and 10^6 in the reduced 6D phase space, with per-trajectory integration diagnostics and Wada topology characterization. Boundary-targeted active learning improves classifier macro-F1 by +3.7% in 2D but degrades it by -0.7% in 6D, where the uncertainty dimension $D_u = 5.86$ causes the fractal boundary to nearly fill the phase space. Under architecture-specific training, measured error-scaling slopes are consistent with the 1983 Grebogi-Ott-Yorke prediction to 6–9% for two architectures. Under a pre-registered matched-compute protocol, slopes are flat; training diagnostics reveal differential convergence as the root cause. Whether classifier scaling tightly matches the GOY bound under properly matched training remains an open question. We release all data and code.

1 Introduction

The gravitational three-body problem is a canonical example of deterministic chaos: the long-term trajectory of three interacting masses is generically unpredictable, yet the asymptotic outcome, which body escapes from the remaining binary, is a well-defined classification problem. The decision boundaries of this classification are the *basin boundaries* of the dynamical system, which are known to be fractal in chaotic scattering systems (Grebogi et al., 1983; McDonald et al., 1985).

A classical result of Grebogi et al. (1983) connects fractal geometry to predictability: for a classifier trained on N uniformly sampled points in d dimensions, the test error scales as $\text{error}(N) \sim N^{-\alpha/d}$, where α is the *uncertainty exponent* measuring how quickly the fraction of ambiguous initial conditions vanishes with perturbation magnitude. This bound has been a theoretical fixture for four decades, but no quantitative test has been performed on a physically grounded system using modern neural classifiers.

We contribute three things. First, a reproducible benchmark dataset on the Montgomery shape sphere (Montgomery, 2015) (2D, 10^6 initial conditions) and in the full 6D reduced phase space (10^6 initial conditions with non-zero momenta), with per-trajectory integration diagnostics. Second, a clean empirical finding: active learning that targets basin boundaries helps in 2D but fails in 6D, explained by the uncertainty dimension. Third, an open methodological problem: classifier error-scaling slopes match the GOY prediction under architecture-specific training but not under matched-compute training, with a mechanistic diagnostic explaining why.

2 Related Work

Trani et al. (2024) mapped the escape-basin structure of the three-body problem across a 2D parameter slice and measured multi-fractal correlation dimensions $D_2 \approx 1.65\text{--}2.0$. Umehara and Tanikawa (1998) studied

binary and triple collisions in the free-fall three-body problem. The fractal-basin framework originates with Grebogi et al. (1983) and McDonald et al. (1985), who introduced the uncertainty exponent and its measurement protocol. Nusse and Yorke (1992) and Poon et al. (1996) established the Wada property for multi-basin chaotic scattering. Aguirre et al. (2001) studied Wada basins in the Hénon-Heiles system. Daza et al. (2016) introduced basin entropy as a complementary diagnostic, and Daza et al. (2018) developed the merging test for the Wada property.

Neural classifiers for basin prediction have been explored by Shena et al. (2021) and Valle et al. (2024) on low-dimensional dynamical systems. Machine learning for the three-body problem specifically has focused on trajectory prediction (Breen et al., 2020; Liao et al., 2022) rather than basin classification. Statistical escape theories (Stone and Leigh, 2019; Manwadkar et al., 2021; Kol, 2021) provide complementary analytical frameworks. Our S_3 -equivariant architecture draws on the DeepSets framework (Zaheer et al., 2017).

3 Benchmark and Setup

Shape sphere. The configuration space of three equal-mass bodies in the plane, modulo translation, rotation, and scaling, is the Montgomery shape sphere S^2 (Montgomery, 2015). A point on this sphere is parameterized by coordinates (n_1, n_2, n_3) with $n_1^2 + n_2^2 + n_3^2 = 1$. Three binary-collision singularities sit 120° apart on the equator; the Lagrange equilateral configurations occupy the poles.

Integration. We integrate the Newtonian equations of motion using a 6th-order Yoshida symplectic integrator (Yoshida, 1990) with adaptive timestep. Trajectories that approach a close encounter below $r_{\min} = 0.01$ length units at any point during integration are excluded, removing 9.1% of the 2D sample and 3.4% of the 6D sample. We classify each surviving trajectory by the Standish escape criterion: a body is considered escaped when it reaches hyperbolic velocity relative to the remaining binary.

Datasets. The 2D at-rest dataset contains 953,143 clean labeled trajectories (after close-encounter exclusion) uniformly sampled on S^2 with zero initial momenta. The 6D dataset contains 1,010,567 clean trajectories with non-zero Jacobi momenta sampled uniformly in the reduced phase space. Both are four-class problems: bound (no escape within the integration window) and three escape outcomes (body 1, 2, or 3). The 2D dataset is approximately 94% bound.

Uncertainty exponent. We measure α via the MGOY algorithm (McDonald et al., 1985): for each perturbation scale ε , we sample 10^5 pairs of initial conditions separated by ε and compute the disagreement fraction $f(\varepsilon)$. A power-law fit $f(\varepsilon) \sim \varepsilon^\alpha$ yields $\alpha_{2D} = 0.259 \pm 0.019$ ($R^2 > 0.999$) and $\alpha_{6D} = 0.145 \pm 0.004$ ($R^2 = 0.999$). The corresponding uncertainty dimensions are $D_u^{2D} = d - \alpha = 1.74$ and $D_u^{6D} = 5.86$.

Wada topology. We apply the merging test of Daza et al. (2018) to the 2D basins: iteratively merging pairs of basin classes and checking whether the basin boundary remains unchanged. All three escape basins share a common boundary, confirming the Wada property. The basin entropy is $S_b = 0.222$ and the basin boundary entropy is $S_{bb} = 0.472$ (Daza et al., 2016).

4 Active Learning in 2D vs 6D

Does boundary-targeted sampling help on fractal-basin classification? We compare two acquisition strategies at matched total labeling budget: *uniform* sampling (i.i.d. from the domain) and *active learning* using k -nearest-neighbor disagreement detection with tangent-plane perturbation (perturbation scale $\sigma = 0.005$ rad in 2D). At each acquisition round, the AL strategy selects points near detected class boundaries; the uniform strategy draws points at random.

Figure 1 shows the results. In 2D, active learning improves macro-F1 by +3.7% at budget $B = 500k$ (from 0.368 to 0.398). In 6D, uniform sampling outperforms active learning at every budget tested, with AL degrading macro-F1 by -0.5% to -1.0% .

The explanation lies in the uncertainty dimension. In 2D, $D_u = 1.74$: the basin boundary is a fractal curve of dimension 1.74 embedded in a 2D space, occupying a vanishing fraction of the domain at finite resolution.

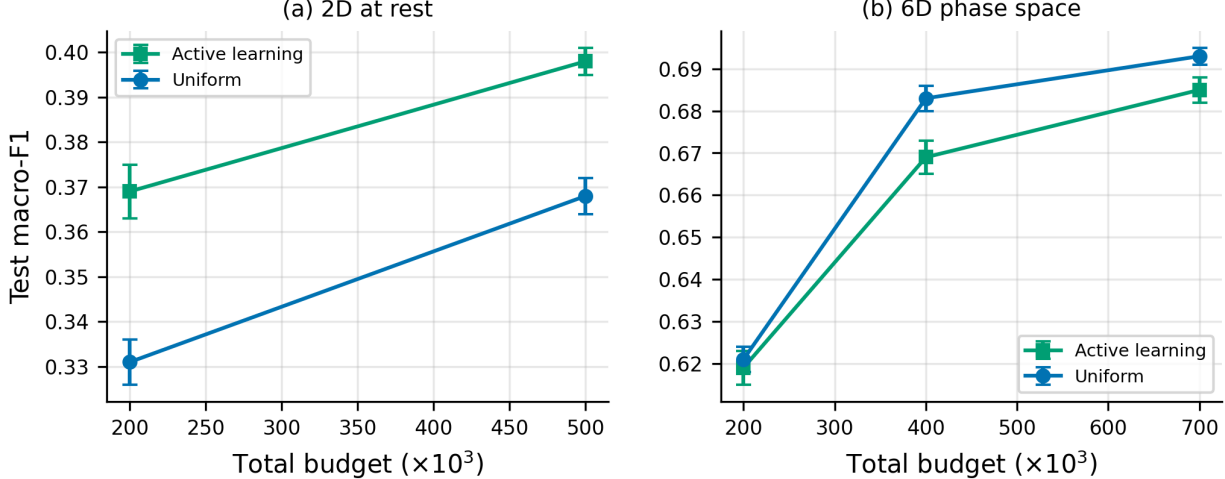


Figure 1: Active learning versus uniform sampling at matched labeling budget. (a) In 2D, AL consistently outperforms uniform sampling. (b) In 6D, uniform sampling performs as well or better than AL at every budget tested. Error bars are 2σ over 5 seeds.

Boundary-targeted sampling concentrates labels where they are most informative. In 6D, $D_u = 5.86$: the boundary nearly fills the 6-dimensional phase space. At any practical resolution, a randomly chosen point is already near a class boundary, so “targeted” sampling degenerates into a noisy approximation of uniform sampling, with the added overhead of k -NN queries. The implication for practitioners is direct: active learning heuristics premised on localized uncertainty fail when uncertainty is spatially uniform, as occurs in high-dimensional fractal-basin systems.

5 The Scaling-Law Question

The GOY bound (Grebogi et al., 1983) predicts that classifier test error on fractal basins scales as $\text{error}(N) \sim N^{-\alpha/d}$. For our 2D dataset with $\alpha = 0.259$ and $d = 2$, the predicted log-log slope is $-\alpha/d = -0.130 \pm 0.010$.

5.1 Architecture-specific training

We train two architectures across $N \in \{10k, 30k, 100k, 300k\}$ with 5 seeds each. The **MLP** is a 6-layer feed-forward network with 384 hidden units, taking 3D shape-sphere coordinates as input, trained for 60 epochs with cosine learning-rate annealing. **S3BodyNet** is a permutation-equivariant architecture (Zaheer et al., 2017) with shared per-body feature extraction and symmetric pooling, taking 7D input (shape-sphere coordinates plus zero-padded Jacobi momenta), trained for 200 epochs under the same optimizer.

Figure 2a shows the results. The MLP produces a slope of -0.131 ± 0.005 , and S3BodyNet produces -0.141 ± 0.009 . Under architecture-specific training, measured slopes are consistent with the GOY prediction to 6–9%.

5.2 Matched-compute protocol

The architecture-specific result confounds training schedule with architecture. An MLP trained for 60 epochs and an S3BodyNet trained for 200 epochs receive different numbers of gradient updates, different learning-rate trajectories, and different convergence characteristics. To isolate the scaling law from training choices, we ran a pre-registered matched-compute test.

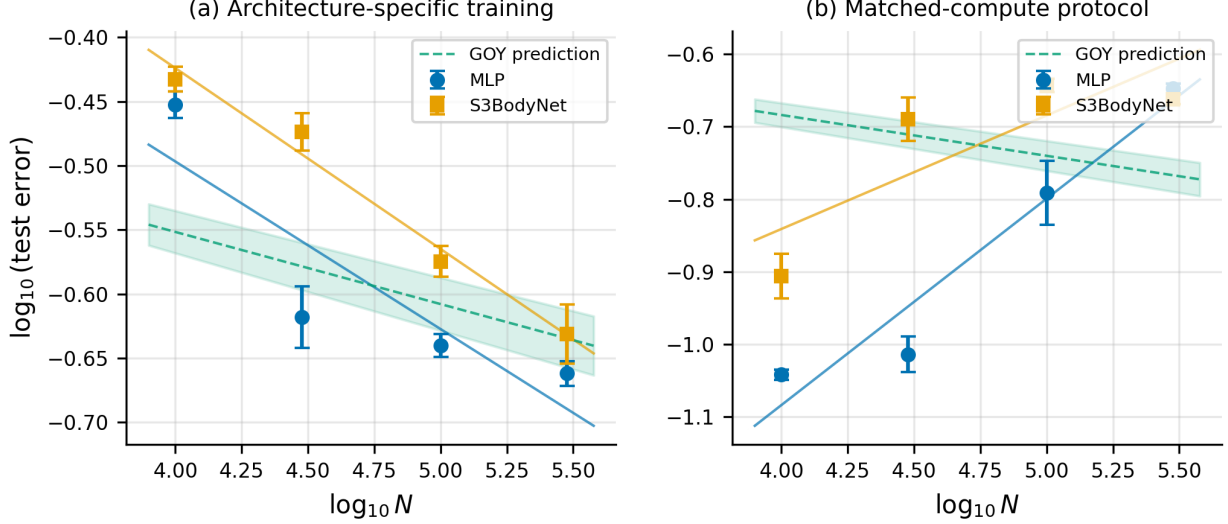


Figure 2: Test error versus training set size on a log-log scale. Green band shows the GOY prediction $\pm 2\sigma$. (a) Under architecture-specific training (MLP at 60 epochs, S3BodyNet at 200 epochs), both slopes fall within the GOY band. (b) Under a matched-compute protocol (identical optimizer, learning rate, and gradient-update budget for both architectures), slopes are flat or inverted, far from the GOY prediction.

The protocol fixes all training hyperparameters across architectures: AdamW optimizer, batch size 1024, a single learning rate (tuned once per architecture on a held-out validation slice), 500-step linear warmup, cosine decay to zero, and a fixed budget of 58,594 gradient updates per cell (equivalent to 200 epochs at $N = 300k$). No early stopping. Five seeds $\{0, 1, 2, 3, 4\}$. Four dataset sizes $N \in \{10k, 30k, 100k, 300k\}$. Pre-registered pass criterion: both slopes within 2σ of the GOY prediction and within 2σ of each other.

Figure 2b shows the results. The MLP slope is -0.002 ± 0.003 ($R^2 = 0.01$), and the S3BodyNet slope is $+0.014 \pm 0.002$ ($R^2 = 0.51$). Both are far from the GOY prediction. The test failed all three agreement criteria.

5.3 Why the disagreement

The fixed gradient-update budget creates a fundamental asymmetry across dataset sizes. At $N = 10k$, 58,594 updates means approximately 6,000 passes through the training data. At $N = 300k$, the same budget provides only 200 passes. We re-trained four representative cells (MLP and S3BodyNet at $N = 10k$ and $N = 300k$) with intermediate logging every 100 gradient updates.

Figure 3 reveals the mechanism. At $N = 10k$, the MLP’s training loss reaches 0.005 (near zero) while validation F1 peaks at 0.418 and declines to 0.388: textbook overfitting on a memorized training set. At $N = 300k$, the MLP’s training loss is still 0.58 at budget end (started at 1.12), with validation F1 still trending upward: the model has not converged. S3BodyNet shows the same pattern more severely, as it converges slower than the MLP on zero-momentum inputs where 5 of 8 per-body features are identically zero.

The opposing biases, overfitting at small N and undertraining at large N , inflate test error at both ends of the sample-size range, producing near-zero slopes. The flat slopes are an artifact of differential convergence, not evidence against the GOY bound.

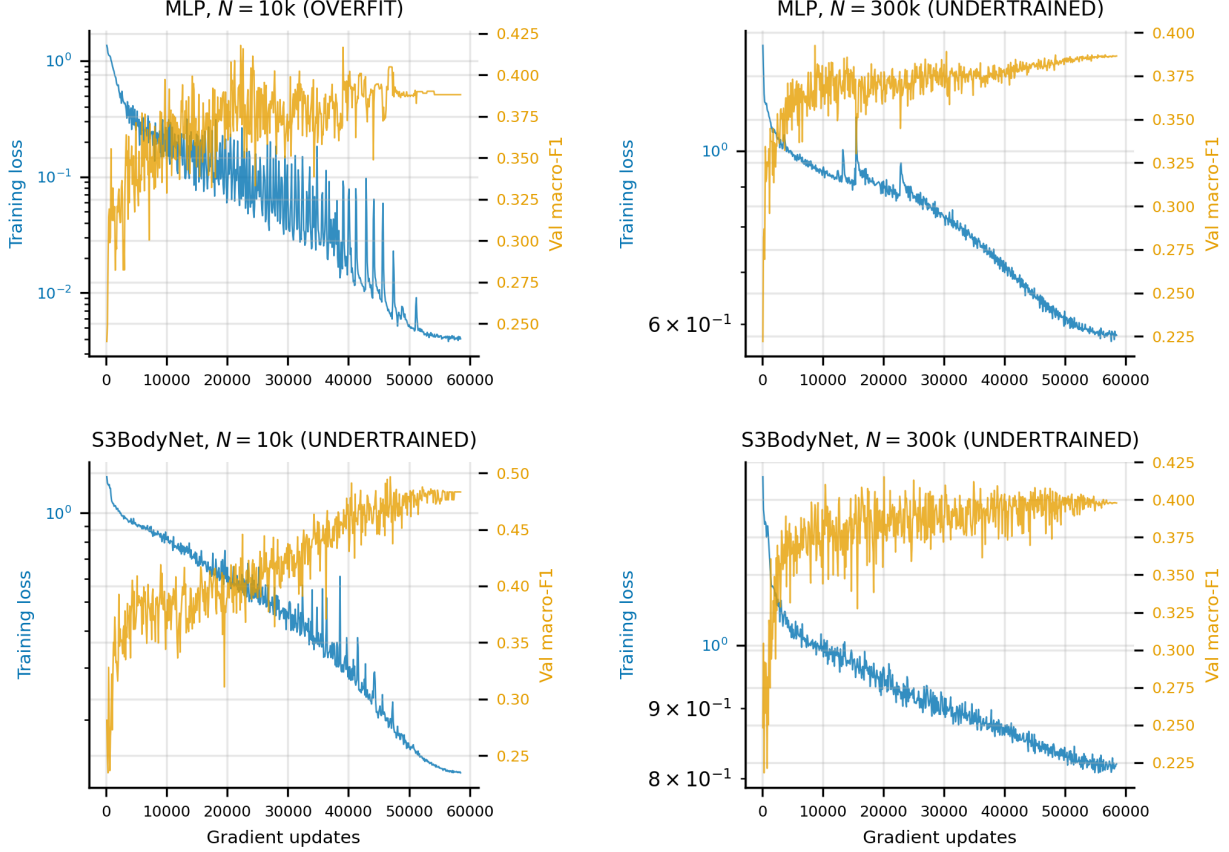


Figure 3: Training diagnostics for four representative cells from the matched-compute protocol. Blue: training loss (log scale). Orange: validation macro-F1. At $N = 10k$, the MLP overfits (training loss near zero, validation F1 declines from peak). At $N = 300k$, both architectures are undertrained (training loss still falling at budget end). This differential convergence flattens the measured slope.

5.4 Interpretation

Neither result alone settles the question. The architecture-specific result is consistent with the GOY prediction but confounds architecture with training schedule. The matched-compute result does not exhibit the predicted scaling but did not achieve equivalent convergence across cells. Whether classifier scaling matches the GOY bound under a protocol that achieves convergence parity across architectures and dataset sizes remains an open question. A future test would need to equalize convergence (for instance, via a train-to-plateau criterion) rather than equalize compute.

6 Discussion

Active learning and dimensionality. The failure of boundary-targeted active learning in 6D is not specific to the three-body problem. Any system whose basin boundary has D_u close to d will exhibit spatially uniform uncertainty, rendering targeted acquisition no better than random sampling. This suggests that active-learning practitioners working on high-dimensional chaotic systems should check D_u before investing in boundary-detection heuristics.

Scaling laws and methodology. The GOY bound is a statement about asymptotic scaling under optimal classifiers. Testing it with finite neural networks introduces confounds, training dynamics, learning-rate

schedules, and convergence rates, that are not part of the theoretical framework. Our results illustrate that bounds from nonlinear dynamics are not trivially testable under standard ML evaluation methodology.

Limitations. This study is restricted to equal-mass bodies in the plane. Unequal masses, post-Newtonian corrections, and energy-conditioned sampling are not covered. The Yoshida-6 integrator without explicit close-encounter regularization is inadequate for hierarchical triples; we address this via conservative close-encounter exclusion (Appendix A). The 2D at-rest slice is a measure-zero subset of the full phase space, though it captures the essential fractal structure.

Future directions. Natural extensions include transfer to other chaotic-scattering benchmarks (Hénon-Heiles), comparison with graph neural networks and equivariant transformer architectures, and a matched-protocol test with convergence-based stopping.

7 Conclusion

We release a benchmark for fractal-basin classification on the planar three-body problem, with 10^6 labeled trajectories in 2D and 6D, Wada topology characterization, and measured uncertainty exponents. Active learning helps in 2D (+3.7%) but fails in 6D (−0.7%), explained by the uncertainty dimension $D_u = 5.86$. Whether classifier error scaling matches the four-decade-old GOY prediction under properly matched training is an open problem; our pre-registered test and its diagnostic provide a concrete starting point. Dataset and code are publicly available.

References

- Jacobo Aguirre, Juan C. Vallejo, and Miguel A. F. Sanjuán. Wada basins and chaotic invariant sets in the Hénon-Heiles system. *Physical Review E*, 64:066208, 2001. doi: 10.1103/PhysRevE.64.066208.
- Philip G. Breen, Christopher N. Foley, Tjarda Boekholt, and Simon Portegies Zwart. Newton versus the machine: solving the chaotic three-body problem using deep neural networks. *Monthly Notices of the Royal Astronomical Society*, 494:2465–2470, 2020. doi: 10.1093/mnras/staa713.
- Alain Chenciner and Richard Montgomery. A remarkable periodic solution of the three-body problem in the case of equal masses. *Annals of Mathematics*, 152:881–901, 2000. doi: 10.2307/2661357.
- Alvar Daza, Alexandre Wagemakers, Bertrand Georgeot, David Guéry-Odelin, and Miguel A. F. Sanjuán. Basin entropy: a new tool to analyze uncertainty in dynamical systems. *Scientific Reports*, 6:31416, 2016. doi: 10.1038/srep31416.
- Alvar Daza, Alexandre Wagemakers, and Miguel A. F. Sanjuán. Ascertaining when a basin is Wada: the merging method. *Scientific Reports*, 8:9954, 2018. doi: 10.1038/s41598-018-28119-0.
- Celso Grebogi, Steven W. McDonald, Edward Ott, and James A. Yorke. Final state sensitivity: an obstruction to predictability. *Physics Letters A*, 99:415–418, 1983. doi: 10.1016/0375-9601(83)90945-3.
- Barak Kol. Flux-based statistical prediction of three-body outcomes. *Celestial Mechanics and Dynamical Astronomy*, 133:17, 2021. doi: 10.1007/s10569-021-10015-x.
- Shijun Liao, Xiaoming Li, and Yu Yang. Three-body problem: from Newton to supercomputer plus machine learning. *New Astronomy*, 96:101850, 2022. doi: 10.1016/j.newast.2022.101850.
- Viraj Manwadkar, Barak Kol, Alessandro A. Trani, and Nathan W. C. Leigh. Chaos and Levy flights in the three-body problem. *Monthly Notices of the Royal Astronomical Society*, 506:692–712, 2021. doi: 10.1093/mnras/stab1689.
- Steven W. McDonald, Celso Grebogi, Edward Ott, and James A. Yorke. Fractal basin boundaries. *Physica D*, 17:125–153, 1985. doi: 10.1016/0167-2789(85)90001-6.

- Richard Montgomery. The three-body problem and the shape sphere. *American Mathematical Monthly*, 122: 299–321, 2015. doi: 10.4169/amer.math.monthly.122.04.299.
- Helena E. Nusse and James A. Yorke. Wada basin boundaries and basin cells. *Communications in Mathematical Physics*, 150:1–21, 1992. doi: 10.1007/BF02096562.
- Leon Poon, Julio Campos, Edward Ott, and Celso Grebogi. Wada basin boundaries in chaotic scattering. *International Journal of Bifurcation and Chaos*, 6:251–265, 1996. doi: 10.1142/S0218127496000035.
- Christos Shena, Constantinos Siettos, and Dimitris Panayotounakos. Using neural networks to predict the basins of attraction in dynamical systems. *Chaos*, 31:023116, 2021. doi: 10.1063/5.0024524.
- Nicholas C. Stone and Nathan W. C. Leigh. A statistical solution to the chaotic, non-hierarchical three-body problem. *Nature*, 576:406–410, 2019. doi: 10.1038/s41586-019-1833-8.
- Alessandro A. Trani, Nathan W. C. Leigh, Tjarda C. N. Boekholt, and Simon Portegies Zwart. Isles of regularity in a sea of chaos amid the gravitational three-body problem. *Astronomy & Astrophysics*, 689: A24, 2024. doi: 10.1051/0004-6361/202449862.
- Hiroaki Umehara and Kiyotaka Tanikawa. Binary and triple collisions causing instability in the free-fall three-body problem. *Celestial Mechanics and Dynamical Astronomy*, 70:167–186, 1998. doi: 10.1023/A: 1026428207880.
- David Valle, Alexandre Wagemakers, and Miguel A. F. Sanjuán. Deep learning-based analysis of basins of attraction. *Chaos*, 34:033105, 2024. doi: 10.1063/5.0159656.
- Haruo Yoshida. Construction of higher order symplectic integrators. *Physics Letters A*, 150:262–268, 1990. doi: 10.1016/0375-9601(90)90092-3.
- Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabás Póczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, 2017.

A Integrator Validation

We use a 6th-order Yoshida symplectic integrator (Yoshida, 1990) with adaptive timestep. Relative energy conservation on the Chenciner-Montgomery figure-eight orbit (Chenciner and Montgomery, 2000) is below 10^{-12} over 100 orbital periods. For general initial conditions, energy is conserved to 10^{-8} or better over the integration window.

Trajectories approaching binary collision ($r_{\min} < 0.01$) experience large force gradients that degrade integration accuracy. Rather than implement explicit regularization, we exclude these trajectories, removing 9.1% of the 2D sample and 3.4% of the 6D sample. This is a conservative choice: excluded trajectories are disproportionately concentrated near the basin boundaries (collision singularities lie on the boundary), so their removal slightly biases the measured uncertainty exponent downward.

B Architecture Details

MLP. 6-layer feed-forward network, 384 hidden units per layer, GELU activation, 5 output units (4 class logits + 1 regression head for log escape time). Input: 3D shape-sphere coordinates (n_1, n_2, n_3) . Trained with AdamW, class-weighted cross-entropy, cosine learning-rate schedule.

S3BodyNet. Permutation-equivariant architecture. Per-body features (8 per body: distances to the other two bodies, pairwise distance products, kinetic energies, center-of-mass distance, radial velocity, angular momentum) are concatenated with global invariants (total energy, total angular momentum, symmetric distance polynomials). A shared 4-layer MLP (256 hidden, 128 output) processes each body’s features. The pooled representation produces the bound logit and regression head; per-body representations produce escape logits. Weight sharing enforces S_3 equivariance by construction.

C Matched-Compute Protocol

The full pre-registered protocol: AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, weight decay 10^{-4} . Batch size 1024. Linear warmup from 0 to peak learning rate over 500 steps, then cosine decay to zero over the remaining budget. Total gradient updates: $\lfloor 200 \times 300,000/1024 \rfloor = 58,594$. Peak learning rate tuned once per architecture on a 20% held-out slice of the $N = 10\text{k}$ training set over a 60-epoch trial: both architectures selected 3×10^{-3} . Five seeds $\{0, 1, 2, 3, 4\}$. No early stopping. Test metric: $1 - \text{macro-F1}$ on a fixed 10^5 held-out test set.

D Four-Way Ablation

At $N = 10^6$ training samples (2D), the ablation results are:

Table 1: Ablation at $N = 10^6$, 2D at-rest dataset, 5 seeds. Macro-F1 on held-out test set.

Architecture	Accuracy	Macro-F1
MLP-raw (3D input)	0.710 ± 0.001	0.699 ± 0.001
MLP-features-noshare	0.711 ± 0.001	0.700 ± 0.001
S3BodyNet (equivariant)	0.662 ± 0.001	0.646 ± 0.001
MLP-raw + S_3 augmentation	0.601 ± 0.001	0.574 ± 0.001

MLP-raw and MLP-features-noshare tie at $N = 10^6$: hand-crafted features provide no additional benefit at this sample size. S3BodyNet’s built-in equivariance does not compensate for its slower convergence on zero-momentum inputs. Data augmentation with S_3 group elements actively degrades performance, consistent with a capacity-dilution effect on data that is already S_3 -symmetric.

E Null and Negative Results

6D scaling law. The GOY prediction for 6D is $-\alpha_{6D}/6 = -0.024$. The measured MLP slope in 6D is approximately -0.054 , a factor of $2.2\times$ steeper. The α_{6D} measurement is clean ($R^2 = 0.999$) and the boundary is not multi-fractal (local α varies by only 0.022 across four decades of ε). The disagreement between predicted and measured slopes in 6D is unexplained.

SIREN architecture. A SIREN network (sinusoidal activations) was tested as an alternative to the GELU-based MLP. Performance was indistinguishable from the baseline MLP at $N = 300k$ and $N = 10^6$. We do not report SIREN results in the main text.